



❖ 1.2.3 MALWARE

Malware, or malicious software, is one of the most common **Cybersecurity** concerns. Malware (malicious software) is any program or file that is harmful to a computer user. Cybercriminals create malware to steal **Passwords**, encrypt files, bypass security systems, or spy on user activities. Hackers use various methods to spread malware and infiltrate computers, smartphones, servers, and **Internet of Things (IoT)** devices. This explains why malware threats are so prevalent, as they are among the most common and dangerous types of cyberattacks.

► **Example:** A student downloads a game from an untrusted website for free. Despite the fact that it installs normally, once installed it includes hidden malware which logs keystrokes and **steals** saved passwords. It can also **spread** to other machines on the same network through shared directories. It would have never happened if the student **had downloaded** this software from a trusted source and **had used** updated antivirus software.



■ How Malware Spreads

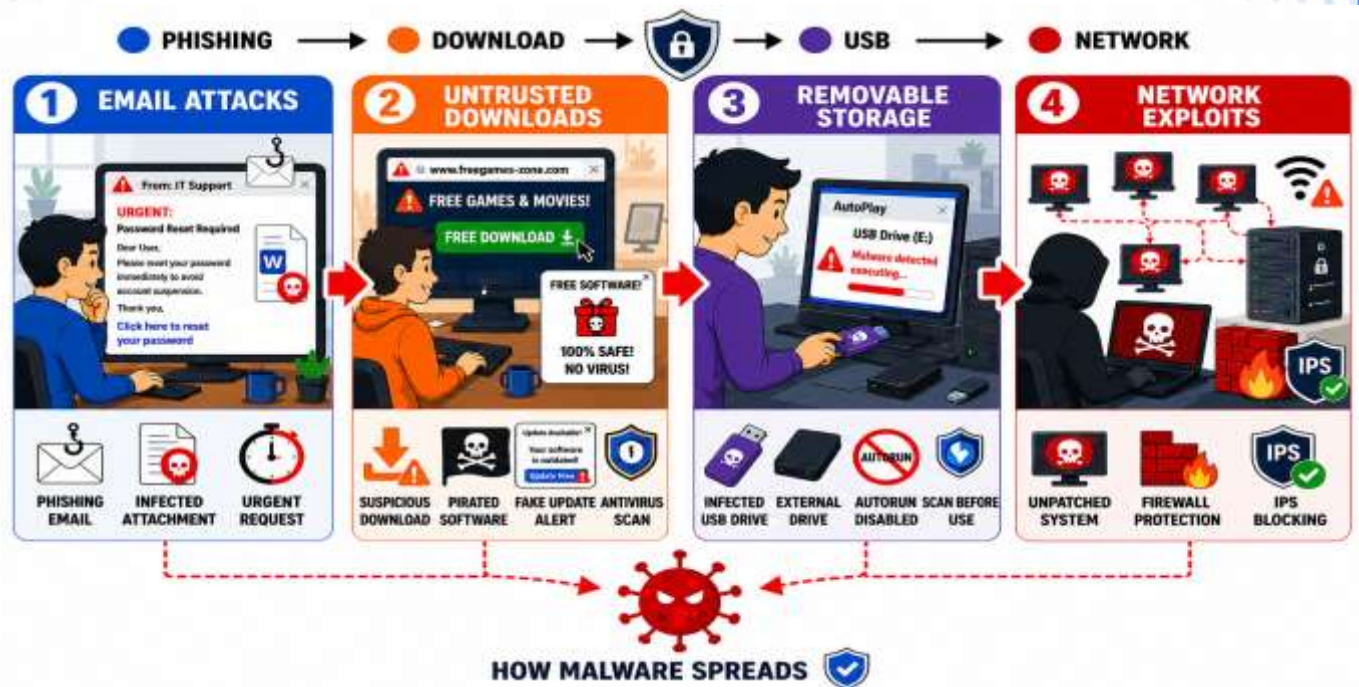
The infection path depends on the **Behavior** of the target and the security weaknesses that an attacker chooses to exploit. The attacker chooses the easiest method of infection depending on the target.

Email Attachments and Phishing: One of the most common delivery methods is **Phishing** emails containing a malicious attachment or link. These emails often impersonate trusted organizations and urge users to open documents or click links immediately.

CHAPTER 2



CYBERSECURITY THREATS & RISKS



Untrustworthy Downloads: Another very common way of becoming infected is downloading programs, games, movies, or documents from unknown or suspicious websites. Similar techniques hide **Malware** inside legitimate programs and software updates.

USB Flash Drives and External Hard Drives: Malicious code can spread to other computers through removable storage devices such as **USB Flash Drives** and external hard drives. The malware may run when malicious files are opened after an infected USB drive is inserted into a computer.

Network / Internet Exploits: Many modern malware types take advantage of software bugs or network weaknesses, including **Zero-Click Attacks**, which require no user interaction. They scan the Internet for outdated systems and exploit critical security vulnerabilities to install malware without informing the victim. These attacks can be prevented by security updates, **Firewalls**, and Intrusion Prevention Systems (IPS).

■ Preventing Malware Infections

Protecting against malware requires a combination of technical **Security Controls** and safe computing practices. Users should install reputable **Antivirus** software and keep it updated so that it can detect new threats as they are discovered. This also includes installing **Security Patches** regularly on operating systems, browsers, and applications where vulnerabilities may exist.

CHAPTER 2



CYBERSECURITY THREATS & RISKS



Users should also use **Firewalls**, email security gateways, and EDR (Endpoint Detection and Response) solutions to reduce malware risks. They should also back up **Data** regularly. Users should avoid opening email attachments from suspicious senders. They should also avoid downloading software or other files from unverified websites and avoid connecting unknown **USB Devices** to their computers. **Cybersecurity Training** helps users understand common malware delivery methods and respond safely.

■ Importance of Malware Protection

Organizations' dependence on digital technology continues to rise, and malware continues to become more **Sophisticated**. Cybercriminals constantly create new **Malware Variants** that attempt to evade traditional security systems. Detecting and containing malware before it spreads requires continuous **Monitoring**, threat intelligence analysis, and incident response planning. Effective **Malware Protection** includes network security, access control, user education, backup strategies, and system monitoring. Organizations can greatly reduce the impact of malware infections when good cybersecurity practices work alongside technology.

❖ 1.2.4 COMMON MALWARE THREATS

The six most common types of **Malware** include viruses, trojans, worms, ransomware, spyware, and adware. Each type has different characteristics, attack methods, and goals. By understanding these differences, users can identify **Cyber Threats** quickly and apply appropriate security controls to protect digital systems.

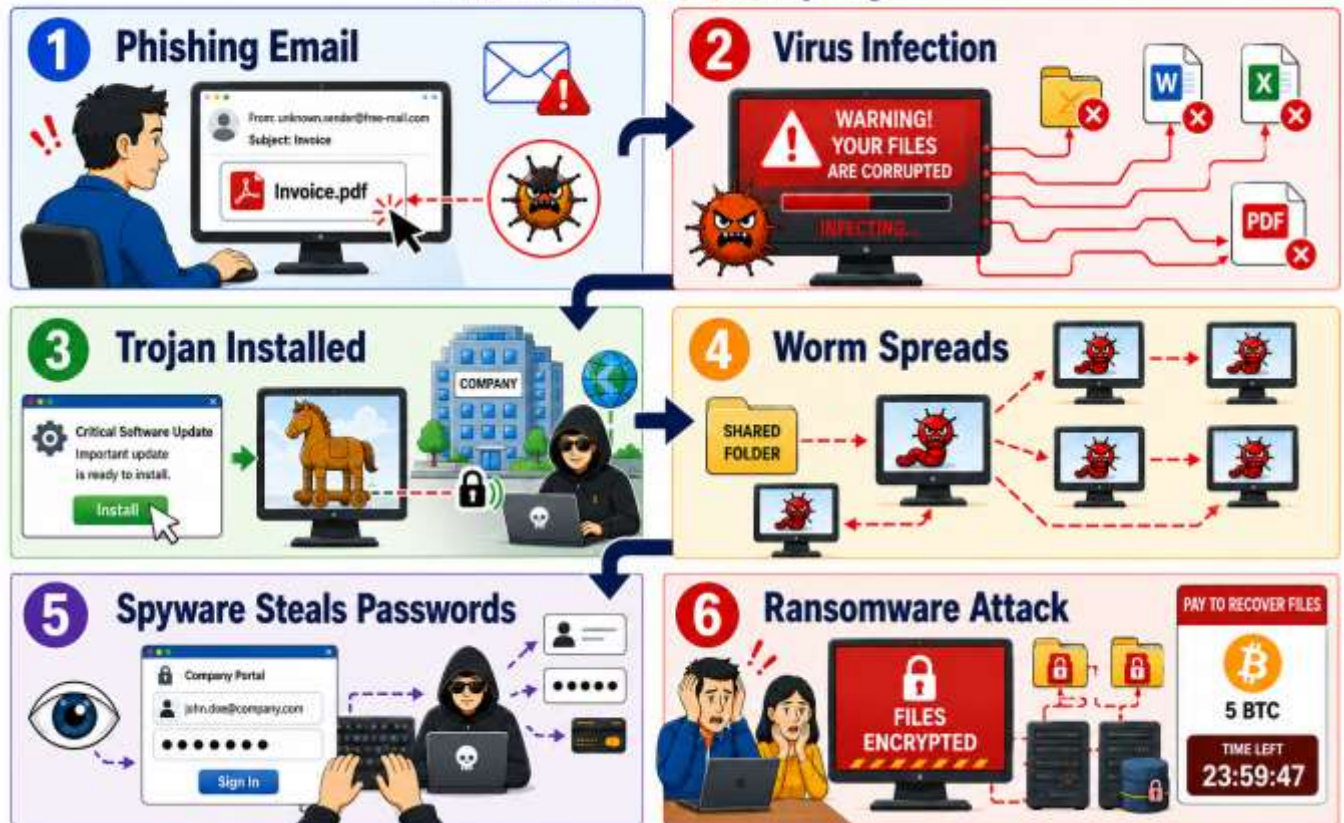
► **Example:** An employee of a company opens an infected email or unknowingly downloads a virus. The virus begins destroying important files on the system. The attackers then install a Trojan disguised as a software update, providing remote access to the company's network. In a matter of minutes, a worm propagates to different computers, spyware is able to stealthily record passwords, and ransomware encrypts shared files. This is an example of a cyberattack where one type of malware works in conjunction with another.

CHAPTER 2



CYBERSECURITY THREATS & RISKS

Attack Inside a Company



■ Protection Against Viruses

A **computer virus** is a type of malicious software (malware) that attaches itself to a legitimate file or program and spreads when the infected file is executed. It can damage files, disrupt system operations, steal information, or infect other computers if left unchecked. Users should install antivirus software, keep it updated, regularly update security, scan USB devices, and avoid downloading suspicious files.

■ Trojan and its Symptoms

A **Trojan Horse** is malware that appears to be useful or a legitimate application. Unlike a **Virus**, a Trojan does not replicate but relies on users to install it. Once installed, it can create a **Backdoor**, steal credit card information or email account passwords, download additional malware, and provide attackers with remote access to the computer. Trojans are commonly disguised as free software, fake antivirus programs, or software updates.

■ Protection against Trojans

When downloading software from an official **Website**, verify digital signatures and do not use cracked software. Companies should also train employees to recognize fake software and fraudulent downloads.

■ Worm and Its Symptoms

A **Worm** is a program that copies itself and spreads to other computers without user intervention. Worms exploit **Network Vulnerabilities** and can infect many devices in an organization within seconds. Slow network performance, unusually high network traffic,

CHAPTER 2



CYBERSECURITY THREATS & RISKS

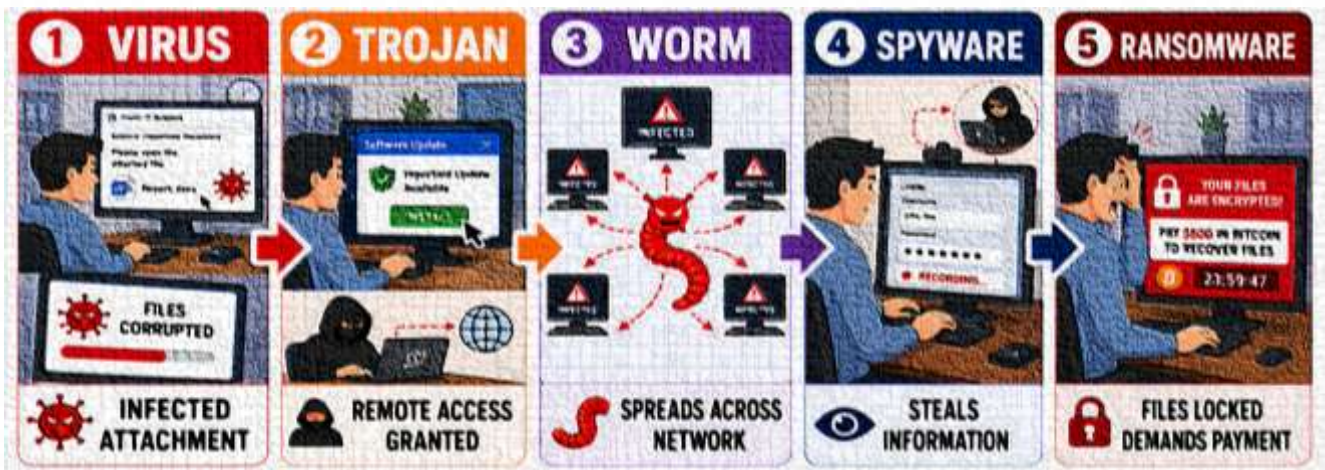
and servers overloaded by infections on multiple computers within a short period may indicate a worm infection.

■ Protection Against Worms

Organizations should apply **Security Updates**, configure **Firewalls** and Intrusion Prevention Systems (IPS), and continuously monitor network traffic whenever vulnerabilities are identified.

■ Ransomware

Ransomware encrypts files so that users cannot access their data unless they pay a ransom. New infections mainly occur through **Phishing** emails, malicious downloads, and software vulnerabilities. The best defense against ransomware is regular **Backups**, email filtering, and security awareness training.



■ Spyware

Spyware secretly collects passwords, banking information, keystrokes, and browsing history. It operates silently, making it difficult for users to detect. Installing quality **Antispyware** software and avoiding suspicious downloads greatly reduces the risk of infection.

■ Adware

Adware is software that automatically displays unwanted advertisements, redirects users' web browsers, and collects personal information for marketing purposes. Some adware may also install additional **Malware** or collect personally identifiable information (PII). Using browser security settings, trusted software sources, and **Antivirus** software helps prevent adware infections.

SESSION REVIEW



1.2.3 MALWARE

How Malware Spreads



- The infection path depends on target behavior and security weaknesses.

Email Attachments and Phishing



- Phishing emails containing malicious attachments or links.

Untrustworthy Downloads



- Downloading programs from unknown or suspicious websites.

USB Flash Drives and External Hard Drives



- Malicious code spreads through removable storage devices.

Network / Internet Exploits



- Software bugs or network weaknesses install malware without user interaction.

Preventing Malware Infections



- Technical security controls and safe computing practices.



Importance of Malware Protection

- Continuous monitoring, user education, and system security reduce malware impact.



1.2.4 COMMON MALWARE THREATS

Protection Against Viruses



- Install antivirus software, update security, and avoid suspicious files.

Trojan and its Symptoms



- Malware appearing to be useful or a legitimate application.

Protection against Trojans



- Download software from official websites and verify digital signatures.

Worm and Its Symptoms



- Program copies itself and spreads without user intervention.

Protection Against Worms



- Apply security updates and continuously monitor network traffic.

Ransomware



- Encrypts files until users pay a ransom.

Spyware



- Secretly collects passwords, keystrokes, and browsing history.

Adware



- Displays unwanted advertisements and collects personal information for marketing.